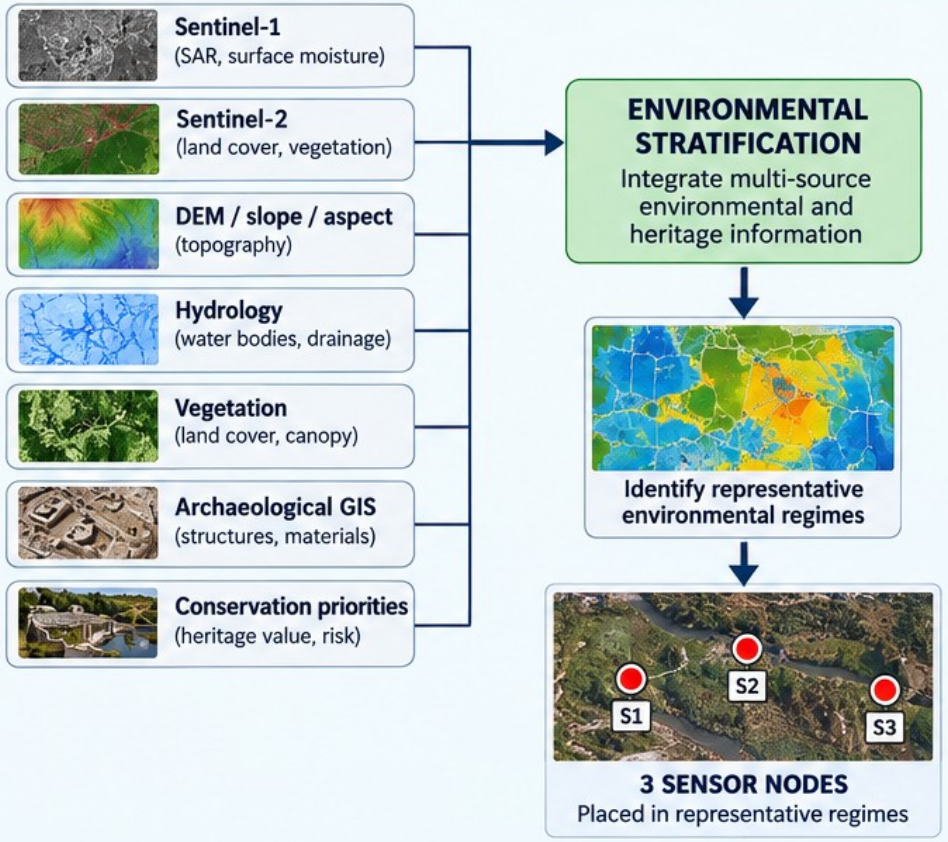
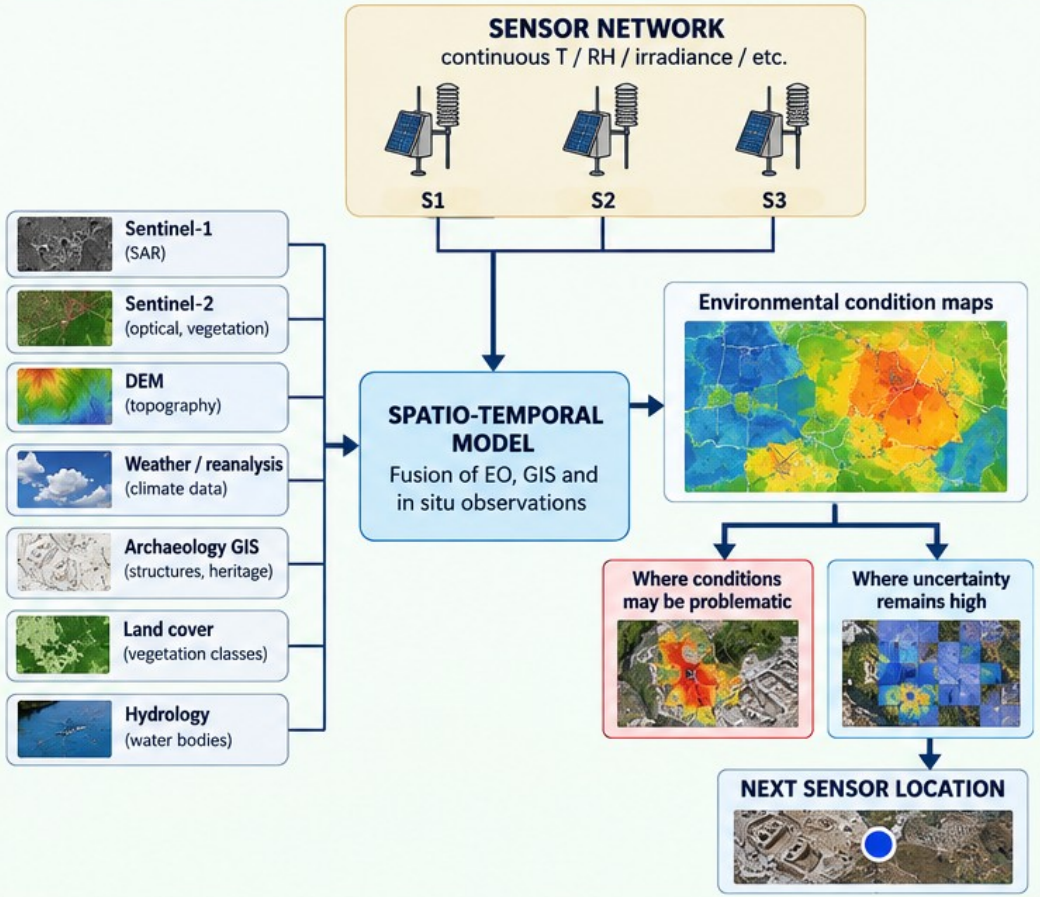
Backup Slides

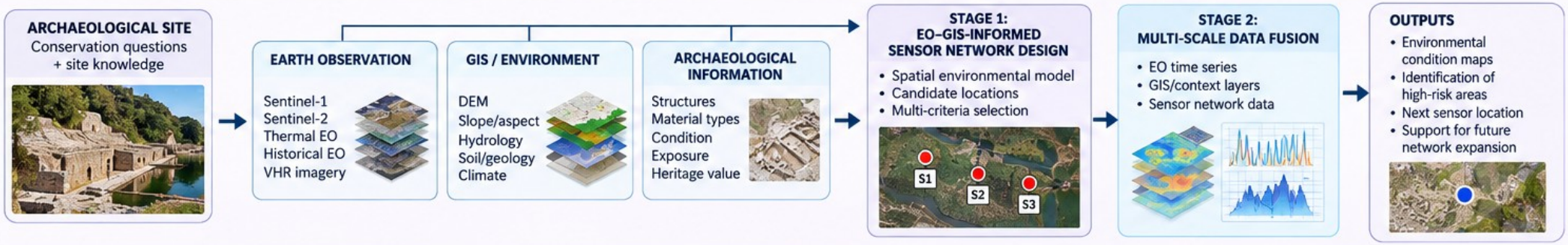
1. Before deployment: Data fusion for optimal node placement



2. After deployment: Data fusion for site monitoring



Two-stage workflow: EO–GIS-informed sensing and data fusion



Three-node monitoring system | Core elements

2 outdoor stations + 1 museum station · Reliable measurements · Expandable LoRaWAN infrastructure

FIELD MEASUREMENT STATIONS

OUTDOOR NODE 01
SOLAR-POWERED FIELD STATION

- Primary T / RH — **Vaisala HMP110**
- Independent check — **Sensirion SHT85**
- Solar exposure — **Apogee irradiance**

POWER Solar + $LiFePO_4$ + MPPT

OUTDOOR NODE 02
SOLAR-POWERED FIELD STATION

- Primary T / RH — **Vaisala HMP110**
- Independent check — **Sensirion SHT85**
- Solar exposure — **Apogee irradiance**

POWER Solar + $LiFePO_4$ + MPPT

MUSEUM NODE 03
INDOOR FIELD STATION

- Primary T / RH — **Vaisala HMP110**
- Independent check — **Sensirion SHT85**
- Light / CO_2 — **Apogee + GMP252**

POWER Mains + battery UPS

AT EVERY NODE **RAK4631 + RAK19001 · RS-485/Modbus, SDI-12, I^2C, analog interfaces**

RTC · Nonvolatile local storage · Watchdog / fault recovery · Power monitoring

COMMUNICATIONS & DATA

TWO LoRaWAN GATEWAYS
Indoor Ethernet + UPS | Outdoor LTE

→

NETWORK SERVER
LoRaWAN LNS → MQTT

→

THINGSBOARD + DATABASE
Storage · QA flags · Visualization

RELIABILITY BY DESIGN Sensor cross-checks · Local buffering · Redundant gateways · ≥1-year unattended target · Scalable to ~20 nodes

Material investigations | Field methods

Targeted, non-destructive surveys complement the permanent environmental monitoring network

TWO COMPLEMENTARY FIELD METHODS

01 SUBSURFACE MATERIAL MOISTURE

Trotec T3000 + TS 610 SDI + TC 30



T3000

Multifunction meter / readout

TS 610 SDI

Microwave moisture probe

TC 30

Sensor connection cable

Indicative moisture mapping up to 30 cm depth

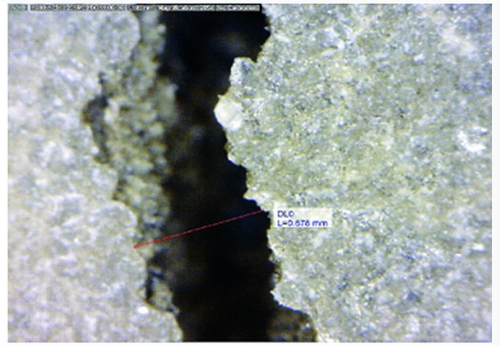
BUDGETED EQUIPMENT **ALL 106,950 (≈107,000)**

02 SURFACE & CRACK INSPECTION

Dino-Lite portable digital microscope



Digital surface imaging



Example crack-width image

Crack measurements and repeatable surface-condition documentation

EXISTING INSTRUMENT

SCIENTIFIC ROLE Map moisture contrasts · Track surface defects · Inform conservation priorities

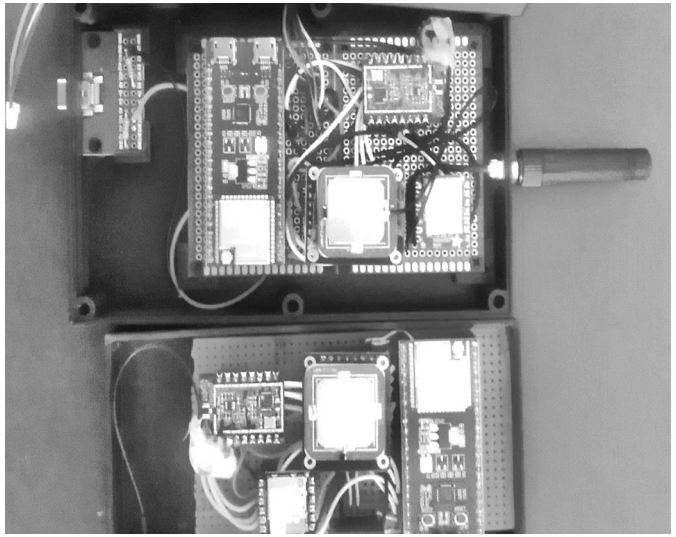
Related ongoing work | Environmental sensing R&D

Existing prototype development demonstrates relevant instrumentation and embedded-systems expertise



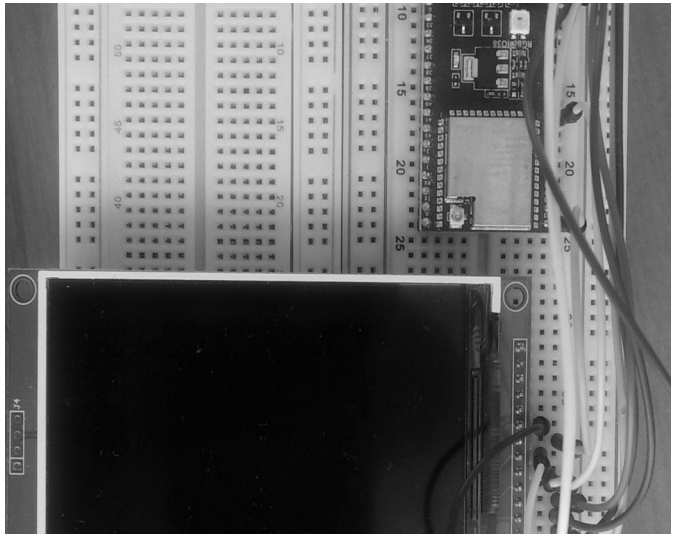
UAV sensing platform

DJI Matrice 350 RTK for mobile environmental sensing



Sensor node prototype

Embedded acquisition, GNSS and LoRa radio modules



Ground interface prototype

ESP32-S3, TFT display and hardware integration

TRANSFERABLE EXPERTISE

Sensor integration · Firmware · GNSS / LoRa · Packaging · Data acquisition

READ
Research Expertise from the Academic Diaspora

Scientific precedents & Butrint context

Selected literature · Earth observation, heritage monitoring and conservation relevance

01 **MT-INSAR**

Deformation mapping

Monitoring Cultural Heritage Sites with Advanced Multi-Temporal InSAR Technique: The Case Study of the Summer Palace

Tang et al. (2016) · Remote Sensing 8(5), 432 · DOI: 10.3390/rs8050432

02 **UAV MULTISPECTRAL**

Archaeological mapping

Investigations of non-visible features in archaeological sites: testing aerial remote sensing with UAV in Pompeii

Santoro et al. (2026) · Virtual Archaeology Review 17(34) · DOI: 10.4995/var.2026.24163

03 **SAR / STRUCTURES**

Masonry monitoring

Assessment of Masonry Wall Deformation through Vertical Displacement Gradients Using COSMO-SkyMed SAR Interferometry: The Case Study of Archaeological Park of Pompeii

Aufiero et al. (2025) · ISPRS Annals X-M-2-2025, 29–37 · DOI: 10.5194/isprs-annals-X-M-2-2025-29-2025

04 **BUTRINT / MATERIALS**

Water, salts, mosaics

Adaptive Conservation Strategies for Butrint's In-Situ Mosaics: Addressing Climate Change Impacts on World Heritage Sites

Nicolini (2025) · Environmental Science & Sustainable Development 10(3) · DOI: 10.21625/essd.v10i3.1173

05 **BUTRINT / POLICY**

Monitoring baseline

Butrint National Park Integrated Management Plan (2020–2030)

Ministry of Culture / AADF (2020) · §11.2 "Monitoring", pp. 174–175 · UNESCO archive

These examples complement—and do not replace—long-term on-site microclimate measurements.

SUPPLEMENTARY REFERENCES